

Refined 3D Index — Full Report

m188

Generated by manifold-index

March 30, 2026

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1 Manifold Overview

Manifold name: **m188**

- Number of ideal tetrahedra: $n = 5$
- Number of cusps: $r = 1$
- Internal edges: $n - r = 4$ (easy: 4, hard: 0)
- Computation order: $q^{20/2}$
- Boundary curve range: $m \in \{-2, -1, 0, 1, 2\}$, $e \in \{-2, -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$

2 Ideal Triangulation (SnaPy Data)

Gluing equation matrix from SnaPy, shape (7×15) .

Edge	(Z_1, Z'_1, Z''_1)	(Z_2, Z'_2, Z''_2)	(Z_3, Z'_3, Z''_3)	(Z_4, Z'_4, Z''_4)	(Z_5, Z'_5, Z''_5)
0	(1, 2, 0)	(1, 0, 2)	(1, 0, 1)	(0, 1, 1)	(0, 0, 0)
1	(0, 0, 1)	(0, 1, 0)	(0, 0, 1)	(0, 0, 1)	(0, 2, 0)
2	(0, 0, 1)	(0, 0, 0)	(0, 1, 0)	(0, 1, 0)	(1, 0, 2)
3	(1, 0, 0)	(1, 0, 0)	(0, 1, 0)	(1, 0, 0)	(0, 0, 0)
4	(0, 0, 0)	(0, 1, 0)	(1, 0, 0)	(1, 0, 0)	(1, 0, 0)

Cusp	(Z_1, Z'_1, Z''_1)	(Z_2, Z'_2, Z''_2)	(Z_3, Z'_3, Z''_3)	(Z_4, Z'_4, Z''_4)	(Z_5, Z'_5, Z''_5)
α	(0, 0, 0)	(0, 0, 0)	(1, -1, 1)	(0, 0, 0)	(0, 1, -1)
β	(-1, 0, 0)	(0, 0, 1)	(-1, 1, 0)	(0, 0, 0)	(0, -1, 1)

3 Edge Classification

Easy edges: 5 total, 4 independent. Hard edges: 0.

Easy Edges			
Index	Independent?	Symbolic Equation	Triplets (f_i, g_i, h_i)
E ₀	✓	$Z_1 + Z_2 + Z'_3 + Z_4 = 2$	(1, 0, 0) (1, 0, 0) (0, 1, 0) (1, 0, 0) (0, 0, 0)
E ₁	✓	$2Z''_3 + 2Z''_4 + 4Z'_5 = 2$	(0, 0, 0) (0, 0, 0) (0, 0, 2) (0, 0, 2) (0, 4, 0)
E ₂	✓	$Z'_2 + Z_3 + Z_4 + Z_5 = 2$	(0, 0, 0) (0, 1, 0) (1, 0, 0) (1, 0, 0) (1, 0, 0)
E ₃	✓	$2Z''_1 + 2Z'_2 = 2$	(0, 0, 2) (0, 2, 0) (0, 0, 0) (0, 0, 0) (0, 0, 0)
E ₄		$Z''_1 + Z'_2 + Z''_3 + Z''_4 + 2Z'_5 = 2$	(0, 0, 1) (0, 1, 0) (0, 0, 1) (0, 0, 1) (0, 2, 0)

4 Neumann–Zagier Data

$g_{\text{NZ}} \in \text{Sp}(2n, \mathbb{Q})$, 10×10 , column ordering $(Z_1, \dots, Z_n, Z''_1, \dots, Z''_n)$.

$$g_{\text{NZ}} = \begin{pmatrix} 0 & 0 & 2 & 0 & -1 & 0 & 0 & 2 & 0 & -2 \\ 1 & 1 & -1 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -4 & 0 & 0 & 2 & 2 & -4 \\ 0 & -1 & 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 2 & -2 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & -1 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 1 \\ 0 & 0 & -1 & 0 & \frac{1}{2} & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \end{pmatrix}$$

Row labels (right margin): $\mu_0, E_0, E_1, E_2, E_3, \lambda_0/2, \Gamma_0, \Gamma_1, \Gamma_2, \Gamma_3$.
Affine shifts: $\nu_x = (0, -1, 2, -1, 0)$, $\nu_p = (0, 0, 0, 0, 0)$.

5 3D Index Formula

The 3D index is computed via:

$$I(\mathbf{m}_{\text{ext}}, \mathbf{e}_{\text{ext}}) = \sum_{\mathbf{e}_{\text{int}} \in (\frac{1}{2}\mathbb{Z})^{n-r}} (-q^{1/2})^{\mathbf{m} \cdot \nu_p - \mathbf{e} \cdot \nu_x} \prod_{a=1}^n I_{\Delta}(m_a, e_a)$$

where $\kappa = (\mathbf{m}_{\text{ext}}, \mathbf{0}^{n-r}, \mathbf{e}_{\text{ext}}, \mathbf{e}_{\text{int}})$ and the local charges are:

$$m_a = (g_{\text{NZ}}^{-1} \kappa)_a, \quad e_a = (g_{\text{NZ}}^{-1} \kappa)_{n+a}$$

Only terms with integer (m_a, e_a) contribute.

Explicit Local Charges. Δ_1 : $m_1 = -2e_{\text{int},3}$, $e_1 = \frac{1}{2}m_0 + e_{\text{int},0}$

$$\Delta_2: m_2 = \frac{1}{2}m_0 + e_{\text{int},2} + 2e_{\text{int},3}, \quad e_2 = e_{\text{int},0} - e_{\text{int},2} - 2e_{\text{int},3}$$

$$\Delta_3: m_3 = -\frac{1}{2}m_0 - 2e_0 + e_{\text{int},0} - 2e_{\text{int},1}, \quad e_3 = m_0 + 2e_0 - e_{\text{int},0} + e_{\text{int},2}$$

$$\Delta_4: m_4 = -2e_{\text{int},1}, \quad e_4 = e_{\text{int},0} + e_{\text{int},2}$$

$$\Delta_5: m_5 = m_0 + 2e_0 + 4e_{\text{int},1}, \quad e_5 = -\frac{1}{2}m_0 - e_0 - 4e_{\text{int},1} + e_{\text{int},2}$$

The tetrahedron index $I_{\Delta}(m, e)$ is given by

$$I_t(m, e) = \sum_{k=\max(0, -e)}^{\infty} \frac{(-1)^k q^{k(k+1)/2 - (2k+e)m/2}}{\prod_{j=1}^k (1 - q^j) \prod_{j=1}^{k+e} (1 - q^j)}$$

with $I_{\Delta}(m, e) = (-q^{1/2})^m \cdot I_t(-m - e, m)$ when $m + e \geq 0$, else $I_{\Delta}(m, e) = I_t(m, e)$.

6 Refined Index Results

Series truncated at $q^{20/2}$ (i.e. $N_{\text{max}} = 10$). Number of computed sectors: 45. Non-zero sectors: 15.

$$\begin{aligned}
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-2)) &= 3q^6 + 5q^{13/2} + 8q^7 + 15q^{15/2} + 23q^8 + 25q^{17/2} + 24q^9 + 20q^{19/2} + 3q^{10} \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-1)) &= q^{5/2} - q^{7/2} - 4q^4 - 7q^{9/2} - 11q^5 - 13q^{11/2} - 11q^6 - q^{13/2} + 16q^7 + 47q^{15/2} + 91q^8 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (0)) &= -q^{1/2} + q + 2q^2 + 3q^{5/2} + 4q^3 - q^{7/2} - 7q^4 - 16q^{9/2} - 27q^5 - 39q^{11/2} - 41q^6 - 3q^7 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (1)) &= q^{3/2} + q^2 + 3q^{5/2} + 2q^3 - q^{7/2} - 8q^4 - 15q^{9/2} - 25q^5 - 34q^{11/2} - 37q^6 - 28q^{13/2} \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (2)) &= q - q^{5/2} - 2q^3 - 4q^{7/2} - 5q^4 - 7q^{9/2} - 8q^5 - 6q^{11/2} + 4q^6 + 17q^{13/2} + 36q^7 + 68q^8 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-2)) &= -3q^{7/2} - 4q^4 - 5q^{9/2} - 6q^5 - 5q^{11/2} + 2q^6 + 11q^{13/2} + 26q^7 + 51q^{15/2} + 84q^8 + 3q^9 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-1)) &= -q + q^{3/2} + 2q^2 + 4q^{5/2} + 3q^3 + q^{7/2} - 6q^4 - 16q^{9/2} - 29q^5 - 41q^{11/2} - 51q^6 - 4q^7 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (0)) &= 1 - 2q^{1/2} - q^{3/2} + 3q^2 + 7q^{5/2} + 11q^3 + 6q^{7/2} - 3q^4 - 17q^{9/2} - 37q^5 - 65q^{11/2} - 51q^6 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (1)) &= -q + q^{3/2} + 2q^2 + 4q^{5/2} + 3q^3 + q^{7/2} - 6q^4 - 16q^{9/2} - 29q^5 - 41q^{11/2} - 51q^6 - 4q^7 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (2)) &= -3q^{7/2} - 4q^4 - 5q^{9/2} - 6q^5 - 5q^{11/2} + 2q^6 + 11q^{13/2} + 26q^7 + 51q^{15/2} + 84q^8 + 3q^9 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-2)) &= q - q^{5/2} - 2q^3 - 4q^{7/2} - 5q^4 - 7q^{9/2} - 8q^5 - 6q^{11/2} + 4q^6 + 17q^{13/2} + 36q^7 + 68q^8 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-1)) &= q^{3/2} + q^2 + 3q^{5/2} + 2q^3 - q^{7/2} - 8q^4 - 15q^{9/2} - 25q^5 - 34q^{11/2} - 37q^6 - 28q^{13/2} \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (0)) &= -q^{1/2} + q + 2q^2 + 3q^{5/2} + 4q^3 - q^{7/2} - 7q^4 - 16q^{9/2} - 27q^5 - 39q^{11/2} - 41q^6 - 3q^7 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (1/2)) &= 0
\end{aligned}$$

7 Weyl Symmetry

Define the Weyl-shifted index:

$$f(\eta_j; m, e) = \eta_j^{a_j \cdot e + b_j \cdot m} \cdot I^{\text{ref}}(m, e)$$

Weyl symmetry requires $f(m, e) = f(-m, -e)$ for each sector pair, where f denotes the Weyl-shifted index above.

Hard edge j	a_j	b_j	$a_j \in \mathbb{Z}?$	$2b_j \in \mathbb{Z}?$
---------------	-------	-------	-----------------------	------------------------

$a \in \mathbb{Z}^k, b \in (\frac{1}{2}\mathbb{Z})^k$ — Dehn filling compatible.

Edge compatibility: each hard edge j must have $a_j \in \mathbb{Z}$; incompatible edges are collapsed ($\eta_j = 1$).

Weyl symmetry $f(m, e) = f(-m, -e)$: 45/45 sectors pass **PASS**

The adjoint $\text{su}(2)$ projection (eq. 2.59–2.61) integrates out η and cusp fugacities against the Haar measure weighted by the adjoint character; the result must equal -1 per filled cusp.

Intermediate c_e values ((q^1, η^0) coefficient of $I^{\text{ref}}(m = 0, e)$):

e	c_e
-2	0
$-\frac{3}{2}$	0
-1	-1
$-\frac{1}{2}$	0
0	0
$\frac{1}{2}$	0
1	-1
$\frac{3}{2}$	0
2	0

Projected value: -1 (expected -1): **PASS**

8 Dehn Filling

8.1 Filling 1: Cusp 0, slope $(0, 1)$

8.1.1 Slope Transformation

- User slope (physical): $P = 0, Q = 1$ (cycle $0 \cdot M + 1 \cdot L$)
- Non-closable cycle: $\gamma = 1 \cdot M + 0 \cdot L$ (slope $1/0$)
- $\text{SL}(2, \mathbb{Z})$ complement: $\delta = 0 \cdot M + 1 \cdot L$
- Verification: $\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$ (should be $+1$)
- User slope in (γ, δ) basis: $(p, q) = (0, 1)$

8.1.2 Hirzebruch–Jung Continued Fraction

$p/q = 0/1$: $[0], \ell = 1$.

8.1.3 g_{NZ} After Basis Change

After replacing cusp basis to the NC cycle $(1, 0)$:

$$g'_{\text{NZ}} = \begin{pmatrix} 0 & 0 & 2 & 0 & -1 & 0 & 0 & 2 & 0 & -2 \\ 1 & 1 & -1 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -4 & 0 & 0 & 2 & 2 & -4 \\ 0 & -1 & 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 2 & -2 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & -1 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 1 \\ 0 & 0 & -1 & 0 & \frac{1}{2} & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \end{pmatrix}$$

$$\nu'_x = (0, -1, 2, -1, 0), \nu'_p = (0, 0, 0, 0, 0).$$

8.1.4 Physical Weyl Vectors

$$\mathbf{a}_{\text{phys}} = ()$$

$$\mathbf{b}_{\text{phys}} = (N/A)$$

All hard edges are compatible ($a_j \in \mathbb{Z}$).

8.1.5 Filled Refined Index

HJ: $[0]$, $\ell = 1$, kernel terms: 35.

$$I_{0/1}^{\text{ref, filled}} = q^{-10} - 2q^{-19/2} + 2q^{-9} - 3q^{-17/2} + q^{-8} - 4q^{-15/2} + 2q^{-7} - 4q^{-13/2} + q^{-6} - 4q^{-11/2} + 4q^{-5} - 4q^{-9/2} + 4q^{-4} - 3q$$